CLOUD COMPUTING - DA - 01 PRIYANKA - G
21MIS1160

1) CLUSTER COMPUTING

→ Cluster computing involves the integration of multiple interconnected computers that function as a single system to enhance computational power, storage and network efficiency.

→ In a leading educational institution, VIT can utilize different types of cluster computing systems to improve academic, research and administrative operations.

→ The adoption of High Performance clusters, load Balancing clusters, storage clusters can streamline data management, support complex research projects and ensure uninterrupted access to online resources.

HIGH PERFORMANCE COMPUTING CLUSTERS

→ Consist of multiple interconnected nodes that enable parallel processing, allowing them to efficiently handle extensive computational work.

APPLICATION AT VIT UNIVERSITY:

→ Supports research in bioinformatics, engineering simulations and AI

→ Facilitates large scale data analysis in fields like genomics and finance.

→ Assist in training deep learning models for applications such as medical diagnostics.

→ Enhance computational fluid dynamics and Finite Element Analysis

IMPLEMENTATION STRATEGY:

→ Deploy GPU and CPU clusters to maximize Parallel processing efficiency.

→ Utilize job scheduling systems such as PBS for workload management

→ Enable remote access portals for faculty and students to efficiently utilize resources

→ Establish dedicated research labs with high-performance nodes and scalable configurations

→ Provide training programs for faculty and students to enhance cluster utilization.

LOAD BALANCING CLUSTER:

→ Distribute computing and network traffic across multiple servers.

→ Applications includes managing peak loads during admission, exam results and course registration.

→ Ensure seamless access to learning management system

→ prevents downtime and provides a smooth experience for large-scale online exams
→ Supports uninterrupted access to digital resources and research databases.

IMPLEMENTATION STRATEGY:
→ Utilize load Balancers such as HA Proxy, Nginx (OR) AWS Elastic Load Balancer.
→ Implement auto-scaling mechanisms to handle varying workloads efficiently.
→ Use container orchestration tools like Kubernetes for dynamic application scaling.
→ Monitor system performance in real time to optimize resource allocation.

STORAGE CLUSTERS:
→ Provide scalable and high availability storage solutions, ensuring efficient data management for academic and research purpose.

APPLICATIONS:
→ Secure storage and management of storage data [student] including enrollment and academic history
→ Centralized storage for research projects handling large datasets.

→ Ensure redundancy and safeguard against data loss.

→ Streamline academic and administrative workflows.

IMPLEMENTATIONS

→ Deploy distributed storage system like Ceph, GlusterFS, OR HDFS.

→ Utilize RAID (Redundant Array of Independent disks) for enhanced redundancy

→ Automate data replication and backup process to improve reliability.

→ Ensure compliance with data security policies through encryption.

→ Implement a tiered storage strategy

By Implementing this, VIT can enhance its research capabilities, optimize administrative operations.

8) PUBLIC, PRIVATE, HYBRID CLOUD

→ Cloud computing has revolutionized the way organisations manage their IT infrastructure by offering scalable, on-demand computing resources.
→ VIT as a leading institution can greatly benefit from cloud computing by enhancing its academic, research and administrative operations.

PUBLIC CLOUD:
→ Public cloud services are offered by third-party providers like Amazon web services, Google Cloud Platform and Microsoft Azure.
→ These services provide scalable computing, storage and networking resources that can be accessed over the Internet on a pay-per use basis.

APPLICATIONS:
→ VIT's LMS can be hosted on a Public Cloud Platforms to perform / support thousands of students and faculty members simultaneously.
→ Cloud hosted virtual labs enable research scholars to collaborate with universities worldwide.
→ Provide coding tools for students to practice programming.
→ Process large datasets for research in fields like

genomics, climate change modeling and AI applications.

IMPLEMENTATION

→ Using cloud based virtual machine and storage for hosting academic applications.
→ Using cloud-based applications like Google workspace, Microsoft 365 for admin
→ Ensure that academic data is securely backed up and protected from accidental loss.

PRIVATE CLOUDS

→ Dedicated infrastructure used exclusively by the university. Provides higher security and control over data and resources.

APPLICATIONS:

→ Student records, faculty payroll and examination results running secure cloud.
→ Hosting internal student portal, research databases and faculty management system on a private cloud enhance security.
→ Providing internal cloud hosted services for security for faculty and students collaboration

IMPLEMENTATIONS:

→ Deploy openstack (or) vmware to manage cloud infrastructure
→ Implement role based Access control to ensure only authorized users can access
→ Use VPN's and Firewalls to protect cloud servers from cyber threats.

HYBRID CLOUDS:

→ Combines both Private and public cloud to create a flexible and scalable IT ecosystem

APPLICATIONS:

→ A hybrid cloud can handle secure storage of question papers on a private cloud while leveraging Public cloud resources to support thousands of student during exams.
→ Research labs can use the private cloud for storing proprietary data and public cloud for additional computing power during peak demand.
→ Data Backups can be stored securely on a Public cloud, ensuring recovery in case of System failures.

COMMUNITY CLOUD

→ Shared cloud infrastructure that serves multiple institutions with common objectives.

APPLICATIONS:

→ Sharing computational resources and datasets with other universities enhances research opportunities.

→ Academic resources, e-books, and research papers can be stored in a community cloud, providing seamless access to students.

→ Government funded research projects can be hosted on a community cloud.

IMPLEMENTATION:

→ Partner with national/ International universities to establish shared cloud.

→ Deploy community cloud platforms like OpenNebula to manage shared resources.